

Work Experience

Meta, FAIR

Research Scientist ([Jason Weston](#)'s team; previously [Kamalika Chaudhuri](#)'s team)

San Francisco, USA

2024 – current

- o Research in AI alignment, reliability, safety, reasoning, self-improvement
- o Contributed to [Muse Spark](#) model and [Muse Spark Safety & Preparedness Report](#)
- o FAIR Better Engineering Award 2026

Princeton University, Visual AI group

Visiting Researcher, supervisor: professor [Olga Russakovsky](#)

Princeton, USA

2024 – 2025

Education

Ph.D. in Machine Learning, New York University

Center for Data Science; supervisor: professor [Andrew Gordon Wilson](#)

New York, USA

2019 – 2024

Thesis: On the Reliability of Deep Learning Models: Uncertainty and Generalization Under Distribution Shifts

Graduate student in Operations Research, Cornell University

Operations Research and Information Engineering department; transferred to NYU

Ithaca, USA

2018 – 2019

B.Sc. in Computer Science, Higher School of Economics

Computer Science department; supervisor: professor [Dmitry Vetrov](#)

Moscow, Russia

2014 – 2018

Cumulative GPA: 9.1 (10.0 scale), class rank: top 3%

Internship Experience

Meta, FAIR

Visiting Researcher (FAIR-NYU AI Mentorship program); mentor: [Mark Ibrahim](#)

2022 – 2024

Research Intern at AI Integrity team; mentors: [Hamed Firooz](#), [Randall Balestriero](#)

June – Sep 2022

Cold Spring Harbor Laboratory

Research Intern; supervisor: professor [Anthony Zador](#)

Cold Spring Harbor, USA

Research topics: meta-learning with compressed neural networks

June – Aug 2021

Google DeepMind

Research Scientist Intern; mentors: [Mehrdad Farajtabar](#), [Razvan Pascanu](#),

[Balaji Lakshminarayanan](#)

Mountain View, USA

June – Oct 2020

Research topic: continual learning with deep generative models

École Polytechnique Fédérale de Lausanne (EPFL)

Machine Learning and Optimization Lab

Lausanne, Switzerland

Research Intern; supervisors: professors [Martin Jaggi](#) and [Dan Alistarh](#)

June – Aug 2018

Research topic: low precision training of neural networks

Google

Software Engineering Intern, Google Cloud Platform Team

Jun – Sep 2017, Seattle, USA

Software Engineering Intern, version control system team

Jun – Oct 2016, Munich, Germany

Publications

* denotes equal contribution or equal advising

Muse Spark Safety & Preparedness Report

[\[link\]](#)

MSL Preparedness & Red Teaming & Alignment Team, AI Security Team; **role: contributor**
preprint

Quantized Reasoning Models Think They Need to Think Longer, but They Do Not

[\[arXiv\]](#)

Sanae Lotfi, **Polina Kirichenko**, Steven Li, Zechun Liu
NeurIPS 2026; oral presentation (top 0.3%)

LLM Knowledge is Brittle: Truthfulness Representations Rely on Superficial Resemblance

Patrick Haller, Mark Ibrahim, **Polina Kirichenko**, Levent Sagun, Samuel J. Bell

[\[arXiv\]](#)

Conference on Language Modeling (COLM) 2026

Visual Compositional Tuning

[\[arXiv\]](#)

Xindi Wu*, Hee Seung Hwang*, **Polina Kirichenko**, Esin Tureci, Olga Russakovsky
International Conference on Learning Representations (ICLR) 2026

AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions

[\[arXiv\]](#), [code](#), [HF](#)

Polina Kirichenko*, Mark Ibrahim*, Kamalika Chaudhuri, Samuel Bell*
Neural Information Processing Systems (NeurIPS) 2025
Used in [GPT-5 system card](#)

What's in Common? Multimodal Models Hallucinate When Reasoning Across Scenes

[\[arXiv\]](#)

Candace Ross, Florian Bordes, Adina Williams, **Polina Kirichenko**, Mark Ibrahim
Neural Information Processing Systems (NeurIPS) 2025

The Impact of Coreset Selection on Spurious Correlations and Group Robustness

[\[arXiv\]](#)

Amaya Dharmasiri, William Yang, **Polina Kirichenko**, Lydia Liu, Olga Russakovsky
Neural Information Processing Systems (NeurIPS) 2025

Position: Out-of-Distribution Detection Methods Answer the Wrong Questions

[\[arXiv\]](#)

Yucen Lily Li, Daohan Lu, **Polina Kirichenko**, Shikai Qiu, Tim Rudner, Bayan Bruss, Andrew Gordon Wilson
International Conference on Machine Learning (ICML) 2025 Position track

Modeling Caption Diversity in Contrastive Vision-Language Pretraining

[\[arXiv\]](#)

Samuel Lavoie, **Polina Kirichenko***, Mark Ibrahim*, Mahmoud Assran, Andrew Gordon Wilson, Aaron Courville, Nicolas Ballas
International Conference on Machine Learning (ICML) 2024

Decomposed Evaluations of Geographic Disparities in Text-to-image Models

[\[arXiv\]](#)

Abhishek Suresddy, Dishant Padalia, Nandhinee Periyakaruppa, Oindrila Saha, Adina Williams, Adriana Romero-Soriano, Megan Richards*, **Polina Kirichenko***, Melissa Hall*
Trustworthy Multi-modal Foundation Models Workshop at ICML 2024; Outstanding paper award

Does Progress On Object Recognition Benchmarks Improve Real-World Generalization?

[\[link\]](#)

Megan Richards, **Polina Kirichenko**, Diane Bouchacourt, Mark Ibrahim
ICML Data-centric Machine Learning Research Workshop 2023
International Conference on Learning Representations (ICLR) 2024

Understanding the Detrimental Class-level Effects of Data Augmentation

[\[arXiv\]](#)

Polina Kirichenko, Mark Ibrahim, Randall Balestrieri, Diane Bouchacourt, Rama Vedantam, Hamed Firooz, Andrew Gordon Wilson
ICML Workshop on Spurious Correlations, Invariance, and Stability 2023
Neural Information Processing Systems (NeurIPS) 2023

Last Layer Re-Training is Sufficient for Robustness to Spurious Correlations

[\[arXiv\]](#), [code](#)

Polina Kirichenko*, Pavel Izmailov*, Andrew Gordon Wilson
ICML Workshop on Spurious Correlations, Invariance, and Stability 2022; oral presentation
International Conference on Learning Representations (ICLR) 2023; spotlight

On Feature Learning in the Presence of Spurious Correlations [\[arXiv, code\]](#)

Pavel Izmailov*, **Polina Kirichenko***, Nate Gruver*, Andrew Gordon Wilson

First presented at *ICML Workshop on Principles of Distribution Shift 2022*

Neural Information Processing Systems (NeurIPS) 2022

Chroma-VAE: Mitigating Shortcut Learning with Generative Classifiers [\[arXiv\]](#)

Wanqian Yang, **Polina Kirichenko**, Micah Goldblum, Andrew Gordon Wilson

Neural Information Processing Systems (NeurIPS) 2022

Does Knowledge Distillation Really Work? [\[arXiv, code\]](#)

Samuel Stanton, Pavel Izmailov, **Polina Kirichenko**, Alexander A. Alemi, Andrew G. Wilson

Neural Information Processing Systems (NeurIPS) 2021

Task-agnostic Continual Learning with Hybrid Probabilistic Models [\[arXiv, poster\]](#)

Polina Kirichenko, Mehrdad Farajtabar, Dushyant Rao, Balaji Lakshminarayanan, Nir Levine, Ang Li, Huiyi Hu, Andrew Gordon Wilson, Razvan Pascanu

*ICML Workshop on Invertible Neural Networks and Normalizing Flows 2021; **spotlight talk***

Why Normalizing Flows Fail to Detect Out-of-Distribution Data [\[arXiv, code\]](#)

Polina Kirichenko*, Pavel Izmailov*, Andrew G. Wilson

First presented at *Workshop on Invertible Neural Networks and Normalizing Flows at ICML 2020*

Neural Information Processing Systems (NeurIPS) 2020

Semi-Supervised Learning with Normalizing Flows [\[arXiv, poster, code\]](#)

Pavel Izmailov*, **Polina Kirichenko***, Marc Finzi*, Andrew G. Wilson

First presented at *Workshop on Invertible Neural Networks and Normalizing Flows at ICML 2019*

International Conference on Machine Learning (ICML) 2020

Subspace Inference for Bayesian Deep Learning [\[arXiv, poster, slides, code\]](#)

Pavel Izmailov*, Wesley Maddox*, **Polina Kirichenko***, Timur Garipov*, Dmitry Vetrov, Andrew G. Wilson

First presented at *Workshop on Uncertainty & Robustness in Deep Learning at ICML 2019; **contributed talk***
Uncertainty in Artificial Intelligence (UAI) 2019

SWALP: Stochastic Weight Averaging in Low Precision Training [\[arXiv, code\]](#)

Guandao Yang, Tianyi Zhang, **Polina Kirichenko**, Junwen Bai, Andrew G. Wilson, Christopher De Sa

International Conference on Machine Learning (ICML) 2019

Awards

FAIR Better Engineering Award	2026
Outstanding paper award at ICML workshop on Trustworthy Foundation Models	2024
DeepMind Fellowship	2019
New York University Center for Data Science Graduate Fellowship	2019
Golden HSE Award	[link] , 2019
HSE Alumni Academic Fellowship	[link] , 2019
Cornell Operations Research and Information Engineering Graduate Fellowship	2018
Ilya Segalovich Scholarship (Yandex)	[link] , 2016, 2017
Google Generation Scholarship EMEA (Google Anita Borg Memorial Scholarship)	[link] , 2015

Talks

“AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions”	
- Cornell Tech Trustworthy AI seminar	2025
- ML Collective	2025
“Addressing robustness to biases in vision foundation models”	
- Invited talk at the ECCV Workshop on Uncertainty Quantification for Computer Vision	2024
“Towards robust and reliable deep learning”	
- Princeton, Visual AI Lab seminar	2023
- FAIR Labs, Meta AI	2023
- Microsoft Research, AI Frontiers labs	2023

- “Distribution shifts in machine learning”
 - Guest lecture at the “Introduction to Data Science” course at NYU 2023
- “Leveraging Large Scale Models for Identifying and Fixing Deep Neural Networks Biases”
 - WiML Un-Workshop at ICML 2023
- “Last Layer Re-Training is Sufficient for Robustness to Spurious Correlations”
 - **Spotlight talk at ICLR** [link], 2023
 - Oral Presentation at the ICML Workshop on Spurious Correlations, Invariance, and Stability [link], 2022
 - Genentech, AI seminar 2022
- “Robustness of Deep Learning Models to Distribution Shift”
 - WiML Un-Workshop at ICML 2022
- “Why Normalizing Flows Fail to Detect Out-of-Distribution Data”
 - ML Collective, Deep Learning: Classics and Trends (with Robin T. Schirrmeister) [slides], 2021
 - Facebook AI Research, Uncertainty team 2021
 - CogSys Talks, Technical University of Denmark [video], 2020
 - Capital One, Machine Learning seminar 2020
 - NeurIPS 2020 [video], 2020
 - INNFP+: Invertible Neural Networks and Normalizing Flows workshop at ICML [video], 2020
- “Applications of normalizing flows: semi-supervised learning, anomaly detection, and continual learning”
 - **Keynote talk at ICML Workshop on Representation Learning for Finance** [video], 2021
- “Does your model know what it doesn’t know?”
 - WiML Un-Workshop at ICML 2021
- “Task-agnostic Continual Learning with Hybrid Probabilistic Models”
 - ICML INNFP+ workshop [video], 2021
- “Continual Learning in Neural Networks”
 - Bayesian Methods Research Group seminar [video], 2021
- “Anomaly detection via Generative Models”
 - ODS DafaFest 2020, Uncertainty Estimation in ML Workshop [video], 2020
- “Uncertainty Estimation in Bayesian Deep Learning”
 - WiML Un-Workshop at ICML 2020
- “Subspace Inference for Bayesian Deep Learning”
 - University of Paris-Saclay, UQSay seminar [link], 2021
 - Uncertainty and Robustness in Deep Learning workshop at ICML [video], 2019
 - Higher School of Economics [video], 2019
- “How do we build neural networks we can trust?”
 - **Broad Institute of MIT and Harvard** [video], 2019

Service

- Workshop Organizing: HAMLETS at NeurIPS 2020 and NeurIPS 2021, **Workshop on Spurious Correlation and Shortcut Learning at ICLR 2025** [link; co-lead organizer], **Workshop on Demographic Diversity in Computer Vision at CVPR 2025** [link; lead organizer]
- Conference Area Chair: TMLR, NeurIPS 2025, ICLR 2026, COLM 2026, NeurIPS 2026
- Conference Reviewing: NeurIPS 2019 (top 400 highest-scoring reviewers), 2020, 2022, 2023; ICLR 2020; ICML 2020 (top 33% reviewer), 2023, 2024; UAI 2020; AISTATS 2021, 2022; CVPR 2025; ICCV 2025
- Workshop Reviewing: NeurIPS 2019 WiML, NeurIPS 2019 BDL, ICML 2020 UDL, ICML 2021 INNFP+, ICML 2021 UDL, NeurIPS 2021 BDL, NeurIPS 2022 DistShift, ICML SCIS 2023, NeurIPS ATTRIB 2023